

Proof that the median minimizes

$$(I) \quad J(y) = \sum_{i=1}^N |y - x_i| \quad \text{w.r.t. } y$$

$$\sum_{i=1}^N \text{sign}(y - x_i) = 0$$

$$\text{sign}(y - x_i) = -1 \rightarrow y < x_i$$

$$\text{sign}(y - x_i) = +1 \rightarrow y > x_i$$

for RHS to be 0, we must have equal no. of
-1s and +1s $\rightarrow y = \text{median}(\{x_i\})$

(II)



$$\text{SAD}(y) = |y - x_1| + |y - x_2| = y - x_1 + x_2 - y = x_2 - x_1$$

Suppose $y \notin [x_1, x_2]$

(a) $y < x_1$

$$\text{SAD}(y) = x_1 - y + x_2 - y = x_1 + x_2 - 2y > x_1 + x_2 - 2x_1 = x_2 - x_1$$

(b) $y > x_2$

$$\text{SAD}(y) = y - x_1 + y - x_2 = 2y - x_1 - x_2 > 2x_2 - x_1 - x_2 = x_2 - x_1$$

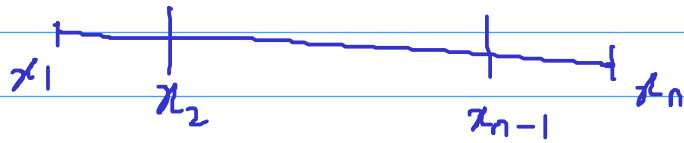
$$\{x_i\} \rightarrow x_1 < x_2 < \dots < x_n$$

$$[x_1, x_n] \quad [x_2, x_{n-1}] \quad \dots \quad [x_i, x_{n+1-i}]$$

where $i = 1$ to C , $C = n/2$ for even n
 $= \frac{n+1}{2}$ for odd n

If n is even, innermost interval = $[x_{\frac{n}{2}}, x_{\frac{n}{2}+1}]$

9/ If n is odd, it is $\left[x_{\frac{n+1}{2}}, x_{\frac{n+1}{2}} \right]$.



In order to minimize SAD, we want y to lie inside every interval.
 $\therefore y$ must lie in the innermost interval.

$$\sum_{i=1}^N |y - x_i| = |y - x_1| + |y - x_5| + |y - x_2| + |y - x_4| + |y - x_3|$$

Proof of 2-sided CE

$\bar{x}$ = sample mean

s = std. dev.

$$\bar{S}_k = \{i \mid |x_i - \bar{x}| < ks\}$$

$$\frac{|\bar{S}_k|}{n} > 1 - \frac{1}{k^2} \quad \left\{ \begin{array}{l} \text{for any } k > 0 \\ \text{unless } k > 1, \text{ the} \\ \text{statement is loose, though} \\ \text{still true)} \end{array} \right.$$

Proof:

$$\begin{aligned} (n-1)s^2 &= \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \sum_{i \in \bar{S}_k} (x_i - \bar{x})^2 + \sum_{i \notin \bar{S}_k} (x_i - \bar{x})^2 \\ &\geq \sum_{i \notin \bar{S}_k} (x_i - \bar{x})^2 \\ &\geq \sum_{i \notin \bar{S}_k} k^2 s^2 = k^2 s^2 (n - |\bar{S}_k|) \end{aligned}$$

$$(n-1)s^2 \geq k^2 s^2 (n - |\bar{S}_k|)$$

$$\frac{n-1}{nk^2} \geq 1 - \frac{|\bar{S}_k|}{n}$$

$$\begin{aligned} \frac{|\bar{S}_k|}{n} &\geq 1 - \frac{n-1}{nk^2} = 1 - \frac{1}{k^2} + \frac{1}{nk^2} \\ &> 1 - \frac{1}{k^2} \end{aligned}$$

$$\frac{|\bar{S}_k|}{n} \leq \frac{1}{k^2}$$